

Minimum Complexity Estimation: Oracle Inequalities, Minimax Bounds and Asymptotic Theory

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June 19, 2026

Abstract

We study a general class of *minimum complexity estimators* (MCE), where the estimator is defined as a penalized likelihood minimizer with a complexity penalty interpreted as a description length or code length. Under a Kraft-type inequality on the penalties, we derive nonasymptotic oracle inequalities for the Hellinger risk (the index of resolvability), minimax lower bounds that match the oracle (index-of-resolvability) rate up to constants, as well as consistency and asymptotic normality under standard regularity conditions. We further pin down the precise minimax status of the procedure: in the parametric regime the estimator is minimax up to constants, whereas on a nonparametric sieve its description-length penalty inflates the rate by a logarithmic factor; we show this factor is a coding artefact, removed exactly—for known smoothness and adaptively—by a penalized-projection refinement that pays the model’s metric complexity rather than the per-coefficient quantization cost. The proofs combine information-theoretic arguments with empirical process techniques and show that the minimum complexity principle yields theoretically sound procedures that adapt to the unknown model complexity.

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1 Introduction

Model selection and prediction under general nonparametric or semiparametric families are among the central problems of modern statistical theory. Since the seminal works of Akaike, Schwarz, Barron, Birgé and Massart, a vast literature has shown that penalized likelihood procedures achieve near-optimal risk bounds under appropriate choices of penalties. More recent contributions, such as van de Geer (2000), have refined these results through local entropy methods and sharp empirical process techniques.

Despite this rich body of work, a persistent limitation remains: *classical oracle inequalities are fundamentally tied to the geometry of the likelihood and to the concentration of the empirical process, rather than to the complexity of the underlying models themselves*. In particular, classical approaches rely on:

- uniform deviation bounds over models indexed by metric complexities (entropy, bracketing, VC dimension, etc.);
- parametric or local asymptotic arguments for well-behaved likelihoods;
- explicitly chosen penalties tuned to the geometry of the model.

In contrast, this paper develops a general theory of *minimum-complexity estimation (MCE)*, in which the selection criterion is driven purely by *code length*—that is, by the amount of information needed to specify the model itself. The central objective is to obtain a unified framework where penalized likelihood, universal coding, and KL-based risk minimization all arise from a single principle:

$$(\widehat{m}, \widehat{\theta}) \in \arg \min_{m, \theta} \left\{ -\log L_n(f_{m, \theta}) + L(m) \right\},$$

with $L(m)$ satisfying Kraft’s inequality. This formulation yields an estimator that is conceptually simple, computationally direct, and theoretically powerful.

Main contributions. We show that the minimum-complexity estimator satisfies a *non-asymptotic, model-wise oracle inequality* for the squared Hellinger risk $h^2(P^*, \hat{f})$ (the Barron–Cover index of resolvability)

$$\mathbb{E} \left[h^2(P^*, \hat{f}) \right] \leq C \left(\inf_{m \in \mathcal{I}} \left\{ \inf_{\theta \in \Theta_m} \text{KL}(P^* \| f_{m,\theta}) + \frac{L(m)}{n} \right\} \right),$$

valid *without metric-entropy, curvature, margin, or sub-model regularity conditions beyond Kraft’s inequality* (for a countable family; a continuous Θ_m requires augmenting $L(m)$ by the code length of a discretization of θ). The Hellinger risk is essential here: the analogous bound for the *Kullback–Leibler* risk does not hold without a further boundedness condition on the log-likelihood ratios, since the penalized maximizer may select a density assigning very small mass where P^* does not, making $\text{KL}(P^* \| \hat{f})$ infinite while the resolvability term stays finite. This is in sharp contrast with the classical theory, where metric-complexity assumptions play an essential role in controlling the empirical process.

The proof rests on a single Kraft/partition-function bound for the Hellinger affinity, rather than bounding the supremum of the empirical process. As a result, no geometry of the parameter space is required beyond the Kraft condition. This provides a form of universality: *any countable model family endowed with prefix code lengths automatically satisfies the oracle inequality.*

We further show that the MCE enjoys:

- a minimax lower bound showing that the rate is essentially optimal for the entire model family;
- strong consistency and asymptotic normality under mild identifiability and regularity, with no need for local bracketing entropy;
- a structural interpretation in terms of universal coding, connecting estimation to information-theoretic principles.

Differences with Barron–Birgé–Massart and van de Geer. Although this work is philosophically aligned with the universal coding approach of Barron and with the penalized likelihood program developed by Birgé and Massart, the present results differ in three fundamental ways:

1. **Elimination of geometric complexity assumptions.** Classical penalties such as AIC, BIC, or MDL rely on model dimensions, local entropy conditions, or the curvature of the log-likelihood. Our bound requires *no such assumptions*: Kraft’s inequality alone suffices, and the empirical process does not need to be controlled on model-dependent neighborhoods.
2. **Uniformity over arbitrarily large model families.** In the Barron–Birgé–Massart framework, the complexity of the model family must be controlled a priori (through entropy or bracketing integrals). Here, the result applies immediately to *any* countable set of models with valid code lengths.

3. **A single structural principle.** Van de Geer’s theory is based on local empirical process methods, margin conditions, and intricate contraction arguments. The present approach derives everything from a canonical likelihood–complexity decomposition and a coding–theoretic argument. The resulting bound is therefore both simpler and fundamentally different: *the estimator is optimal because its description length is optimal.*

Thus, the contribution of this article is not merely to provide another oracle inequality, but to show that oracle optimality can be achieved in full generality from a single information–theoretic principle, with no geometric assumptions on the underlying models. This aligns statistical estimation with a universal coding perspective, offering a clean, minimalistic and powerful foundation for general model selection.

2 Framework and Minimum Complexity Estimator

Let $(\mathcal{X}, \mathcal{A})$ be a measurable space endowed with a σ -finite dominating measure μ . The true distribution $P^\star$ is assumed to admit a density $f^\star$ with respect to μ .

We consider a countable index set $\mathcal{I}$ and, for each $m \in \mathcal{I}$, a model $\mathcal{M}_m$ consisting of densities $f_{m,\theta}$ with $\theta \in \Theta_m$:

$$\mathcal{M}_m = \{f_{m,\theta} : \theta \in \Theta_m\}.$$

We let $\mathcal{M} = \bigcup_{m \in \mathcal{I}} \mathcal{M}_m$ denote the full model family.

Given an i.i.d. sample $(X_1, \dots, X_n)$ distributed according to $P^\star$, the log–likelihood of $f_{m,\theta}$ is

$$\ell_n(m, \theta) := \sum_{i=1}^n \log f_{m,\theta}(X_i).$$

2.1 Complexity penalties and Kraft inequality

To encode the model index m we consider a *code length* function $L : \mathcal{I} \rightarrow [0, \infty)$ satisfying the Kraft inequality

$$\sum_{m \in \mathcal{I}} 2^{-L(m)} \leq 1. \quad (2.1)$$

This ensures the existence of a prefix code with code lengths $L(m)$, and provides a natural information-theoretic interpretation of the penalty.

Typical choices include, for example,

$$L(m) \asymp \dim(\Theta_m) \log n$$

in parametric or sieve-type settings, though our results do not require any specific form as long as (2.1) holds.

2.2 Definition of the minimum complexity estimator

The minimum complexity estimator (MCE) is defined as any measurable selection

$$(\widehat{m}, \widehat{\theta}) \in \arg \min_{m \in \mathcal{I}, \theta \in \Theta_m} \left\{ -\ell_n(m, \theta) + L(m) \right\}. \quad (2.2)$$

We then write

$$\hat{f} := f_{\hat{m}, \hat{\theta}}$$

for the estimated density.

Remark 2.1. *The penalty is placed solely on the model index m . In many practical situations one can refine this to include a parametric complexity term depending on θ but, for clarity, we focus on the purely model-based penalty here.*

3 Main Results

In this section we state the main theoretical results concerning the risk, minimax properties, and asymptotic behavior of the MCE.

3.1 Oracle inequality

Our first theorem is a nonasymptotic oracle inequality for the Kullback–Leibler risk.

Theorem 3.1 (Oracle inequality for the MCE). *Assume that $(L(m))_{m \in \mathcal{I}}$ satisfies the Kraft inequality (2.1). Let $\hat{f}$ be any MCE defined by (2.2). Then there exists a universal constant $C > 0$ such that, for the squared Hellinger distance $h^2(P, Q) = \frac{1}{2} \int (\sqrt{dP} - \sqrt{dQ})^2$,*

$$\mathbb{E} \left[h^2(P^*, \hat{f}) \right] \leq C \inf_{m \in \mathcal{I}} \left\{ \inf_{\theta \in \Theta_m} \text{KL}(P^* \parallel f_{m, \theta}) + \frac{L(m)}{n} \right\}. \quad (3.1)$$

(For a continuous Θ_m , the penalty $L(m)$ must be augmented by the code length of a discretization of θ ; see the proof.)

Remark 3.2. *The inequality (3.1) shows that the MCE automatically adapts to the best model in the collection, up to a complexity term $L(m)/n$ that reflects the description length of the model index. In particular, if P^* is well approximated by a low-complexity model, the risk of the MCE will be correspondingly low.*

3.2 Minimax lower bound

We complement the oracle inequality with a minimax lower bound showing that the oracle rate cannot be improved in a distribution-free sense (up to constants).

Theorem 3.3 (Minimax lower bound). *Let $\mathcal{F}$ be a model class containing a finite subset $\{f_0, \dots, f_M\}$ such that*

1. $\text{KL}(f_j \parallel f_k) \leq \alpha$ for all $j \neq k$;
2. $\text{KL}(f_j \parallel f_0) \geq \beta$ for all $j \geq 1$,

for some $\alpha, \beta > 0$. Then, for any estimator $\hat{f}$ based on n observations,

$$\sup_{f \in \mathcal{F}} \mathbb{E}_f [\text{KL}(f \parallel \hat{f})] \geq c \beta \left(1 - \frac{n\alpha + \log 2}{\log M} \right), \quad (3.2)$$

for a universal constant $c > 0$.

Remark 3.4. *Theorem 3.3 is a standard Fano-type statement. In concrete settings, one can construct finite subsets $\{f_j\}$ with pairwise Kullback–Leibler distances controlled as in (i)–(ii) and thereby obtain explicit lower bounds in terms of n , the dimension of the model, or its metric entropy. The result shows that the rate appearing in the oracle inequality is essentially optimal. The precise gap between this oracle (resolvability) rate and the exact minimax rate on nonparametric sieves—a logarithmic factor of purely coding origin—is analyzed in Section 3.3.*

3.3 From near-minimax to exact: coding cost versus metric complexity

The oracle inequality (3.1) is sharp as a bound on the index of resolvability. When instantiated on a nonparametric sieve, however, the description-length penalty $L(m)$ carries a logarithmic factor that is *absent* from the minimax rate. We isolate this factor, show it is a coding artefact, and remove it.

Consider the canonical sieve: $\mathcal{M}_k$ is a linear space of dimension $D_k \asymp 2^{kd}$ (e.g. piecewise polynomials of degree $< \lceil s \rceil$ at resolution 2^{-k}) approximating a bounded density f^* with $0 < c_0 \leq f^* \leq C_1 < \infty$ and Hölder smoothness s on a d -dimensional domain, so that $\inf_{g \in \mathcal{M}_k} \text{KL}(P^* \| g) \leq C L^2 2^{-2ks}$.

Proposition 3.5 (The MCE rate carries a logarithmic factor). *To obtain a countable family the coefficients of $\mathcal{M}_k$ must be quantized on a grid of spacing δ_k ; with $\delta_k \asymp 2^{-ks}$ this costs $L(k) \asymp D_k \log_2(1/\delta_k) \asymp D_k ks \lesssim D_k \log n$ bits. The oracle inequality (3.1) then gives*

$$\mathbb{E}[h^2(P^*, \hat{f})] \leq C \inf_k \left\{ L^2 2^{-2ks} + \frac{D_k \log n}{n} \right\} \leq C' \left(\frac{\log n}{n} \right)^{2s/(2s+d)},$$

i.e. the minimax rate $n^{-2s/(2s+d)}$ up to a logarithmic factor.

The $\log n$ is thus a *coding artefact*: it is the per-coefficient cost $\log(1/\delta_k)$ of describing a continuum by a prefix code, not a feature of the estimation problem. An estimator that fits the coefficients of $\mathcal{M}_k$ *without* quantizing them pays instead the metric complexity D_k/n of the model and attains the exact rate.

Theorem 3.6 (Exact minimax rate via penalized projection). *Let $\hat{g}_k = \Pi_k P_n$ be the orthogonal $L^2(\mu)$ -projection of the empirical measure onto $\mathcal{M}_k$, and let $\hat{f}$ be its truncation to $[c_0/2, 2C_1]$, renormalized to a density.*

1. (Known s .) *The single resolution $D_{k_n} \asymp n^{d/(2s+d)}$ gives $\mathbb{E} h^2(P^*, \hat{f}_{k_n}) \leq C(2^{-2k_n s} + D_{k_n}/n) \leq C' n^{-2s/(2s+d)}$.*
2. (Adaptive.) *The penalized selection $\hat{k} \in \arg \min_k \{-\|\hat{g}_k\|_2^2 + \kappa D_k/n\}$ attains $\mathbb{E} h^2(P^*, \hat{f}) \leq C \inf_k \{L^2 2^{-2ks} + D_k/n\} \leq C' n^{-2s/(2s+d)}$, without knowledge of s .*

In both cases there is no logarithmic factor.

Proof sketch. (i) Orthogonality of the projection splits $\mathbb{E}\|\hat{g}_k - f^*\|_2^2$ into the approximation term $\|f^* - \Pi_k f^*\|_2^2 \leq CL^2 2^{-2ks}$ and the variance $n^{-1} \sum_{\ell \leq D_k} \text{Var}(e_\ell(X)) \leq C_1 D_k/n$, where (e_ℓ) is an $L^2(\mu)$ -orthonormal basis of $\mathcal{M}_k$ and $\sum_\ell \|e_\ell\|_2^2 = D_k$. The truncation is an L^2 -contraction toward an interval containing f^* , and under the two-sided bound $f^*, \hat{f} \in [c_0/2, 2C_1]$ one has $h^2 \lesssim \|\cdot\|_2^2$; balancing $2^{-2k_n s} \asymp D_{k_n}/n$ gives $n^{-2s/(2s+d)}$. (ii) The criterion is the least-squares contrast $\gamma_n(g) = \|g\|_2^2 - 2P_n g$, minimized over $\mathcal{M}_k$ by $\hat{g}_k$. The Birgé–Massart penalized-projection oracle inequality (Birgé and Massart, 1998; Massart, 2007) applies: since the dimensions $D_k \asymp 2^{kd}$ grow *geometrically*, the weights $x_k \asymp D_k$ are summable ($\sum_k e^{-D_k} < \infty$), so the penalty $\text{pen}(k) \asymp D_k/n$ —proportional to the metric complexity, with no extra log—is admissible and yields $\mathbb{E}\|\hat{g}_k - f^*\|_2^2 \leq C \inf_k \{\|f^* - \Pi_k f^*\|_2^2 + D_k/n\}$. Inserting the approximation bound gives the exact adaptive rate. $\square$

Remark 3.7 (Honest minimax status of the MCE). *Theorem 3.1 shows the MCE is oracle-optimal for the index of resolvability, and Theorem 3.3 shows this index is optimal up to constants. But on a nonparametric sieve the resolvability rate itself exceeds the minimax rate by the coding factor $\log n$ (Proposition 3.5); the MCE is then near-minimax (optimal up to a logarithmic factor), not minimax up to constants. The exact rate is recovered by the penalized-projection estimator of Theorem 3.6, which differs from the MCE precisely by replacing the description-length penalty $D_k \log n$ with the metric-complexity penalty D_k . In the parametric regime (finite-dimensional Θ_m , bounded $L(m)$) no such factor arises and the MCE is minimax up to constants. The same dichotomy, instantiated on a G -invariant density on a quotient E/G of effective dimension $d_{\text{eff}} = \dim(E/G)$, underlies the orbital minimum-complexity estimator: there too the description-length code is near-minimax and the projection estimator is exact.*

3.4 Consistency and asymptotic normality

We now state the main asymptotic results for the MCE.

Theorem 3.8 (Strong consistency). *Assume that:*

1. *The true distribution P^* belongs to $\bigcup_{m \in \mathcal{I}} \mathcal{M}_m$ and is uniquely represented as f_{m^*, θ^*} ;*
2. *For each m , the parameter space Θ_m is compact and the map $\theta \mapsto f_{m, \theta}$ is continuous in KL;*
3. *The penalty satisfies $L(m) = o(n)$ as $n \rightarrow \infty$ for each fixed m and grows sufficiently fast in m to exclude overly complex models.*

Then

$$\hat{m} \rightarrow m^*, \quad \hat{\theta} \rightarrow \theta^*$$

almost surely as $n \rightarrow \infty$.

Under additional smoothness assumptions we obtain asymptotic normality.

Theorem 3.9 (Asymptotic normality). *In addition to the hypotheses of Theorem 3.8, assume that the model $\mathcal{M}_{m^*}$ is regular in the sense of classical likelihood theory, with Fisher information matrix $I(\theta^*)$ positive definite. Then*

$$\sqrt{n}(\hat{\theta} - \theta^*) \xrightarrow{d} \mathcal{N}(0, I(\theta^*)^{-1}).$$

4 Nonasymptotic Tools and Empirical Processes

We briefly sketch the main probabilistic ingredients used in the proofs.

First, we exploit a basic decomposition of the Kullback–Leibler risk in terms of likelihood ratios (Lemma 7.1 in Section 7) combined with the optimality condition (2.2) to compare $\hat{f}$ to any fixed $f_{m,\theta}$. This yields an inequality of the form

$$\text{KL}(P^\star \parallel \hat{f}) \lesssim \text{KL}(P^\star \parallel f_{m,\theta}) + \frac{L(\hat{m}) - L(m)}{n} + \text{empirical process fluctuations},$$

which is then controlled via exponential inequalities for empirical processes indexed by log-likelihood differences.

Second, we use simple Bernstein-type inequalities and, when necessary, entropy bounds for function classes to control the suprema of empirical processes (Lemma 7.3 and Lemmas 7.9–7.10 in Section 7). This allows us to obtain high-probability versions of the oracle inequality.

Third, for the minimax lower bound we rely on a classical Fano argument (see, e.g., Tsybakov, 2009). The key idea is to embed a finite subset of well-separated models into $\mathcal{M}$ and relate the estimation problem to a multi-hypothesis testing problem.

Finally, the consistency and asymptotic normality results follow the standard M-estimation scheme developed in van der Vaart (1998), after verifying that the penalized criterion behaves like the negative log-likelihood in a neighborhood of $(m^\star, \theta^\star)$ and that the penalty does not perturb the local quadratic approximation.

5 Simulation Illustration (Optional)

In a full version of the paper, this section would contain simulation experiments comparing the MCE with other model selection criteria such as AIC, BIC or cross-validation in various parametric and semi-parametric settings. The aim would be to illustrate:

- Adaptation of the MCE to the true model complexity;
- Robustness of the procedure under mild misspecification;
- Empirical verification of the oracle-type behavior suggested by Theorem 3.1.

6 Discussion

We have shown that the minimum complexity principle, when formalized as a penalized likelihood procedure with code-length penalties, enjoys strong statistical guarantees. The resulting estimator achieves oracle-like performance for the index of resolvability, is minimax optimal up to constants in the parametric regime—and up to a logarithmic factor on nonparametric sieves, a coding cost removable by the penalized-projection refinement of Section 3.3—and behaves like a regular M-estimator within the true model.

Several extensions are natural. One direction is to consider data-dependent or hierarchical penalties, possibly reflecting *twin* structures or group actions in the underlying models.

Another direction is to strengthen the nonasymptotic analysis using refined chaining bounds and local Rademacher complexities, which may yield sharper constants and allow for heavier-tailed observations.

Finally, the information-theoretic viewpoint suggests a deeper connection between minimum complexity estimation, universal source coding and the geometry of model spaces—an aspect that is especially relevant in the broader Twin Spaces program.

7 Proofs

In this section we provide complete proofs of the main results stated in Sections 2–3. Throughout, we work on the same probability space and use the notation introduced in the body of the paper. For concreteness, we denote by P^* the unknown data-generating distribution, by $(\mathcal{M}_m)_{m \in \mathcal{I}}$ the model family, and by $f_{m,\theta}$ the densities (or likelihoods) with respect to a dominating measure μ . The *minimum complexity estimator* (MCE) is written $\hat{f} = f_{\hat{m},\hat{\theta}}$.

7.1 Preliminary inequalities

We begin with two elementary lemmas that will be used repeatedly.

Lemma 7.1 (Empirical/expected log-likelihood decomposition). *Let P^* have density f^* and let Q be any probability measure with density q (both with respect to μ). Set $h_Q := \log(f^*/q)$. Then*

$$P^*h_Q = \text{KL}(P^* \| Q), \quad P_n h_Q = \text{KL}(P^* \| Q) + (P_n - P^*)h_Q, \quad (7.1)$$

where $P_n = n^{-1} \sum_{i=1}^n \delta_{X_i}$. Thus the empirical log-likelihood ratio equals the Kullback–Leibler divergence plus a mean-zero empirical-process fluctuation $(P_n - P^*)h_Q$.

Proof. By definition $P^*h_Q = \int f^* \log(f^*/q) d\mu = \text{KL}(P^* \| Q)$. The second identity is the trivial decomposition $P_n h_Q = P^*h_Q + (P_n - P^*)h_Q$, and $\mathbb{E}_{P^*}[(P_n - P^*)h_Q] = 0$. $\square$

Remark 7.2. *It is precisely the fluctuation $(P_n - P^*)h_{\hat{f}}$, evaluated at the data-dependent $\hat{f}$, that must be controlled: the in-sample average $P_n h_{\hat{f}}$ is not equal in expectation to $\mathbb{E}[\text{KL}(P^* \| \hat{f})]$, because $\hat{f}$ is selected using the same sample. The proof of Theorem 7.4 below handles this through a Kraft/partition-function bound on the Hellinger affinity rather than through such a (false) in-sample identity.*

Lemma 7.3 (Exponential inequality for empirical processes). *Let $(X_i)_{i=1}^n$ be i.i.d. with law P^* and let $\mathcal{F}$ be a countable class of measurable functions such that $\|f\|_\infty \leq b$ and $P^*(f^2) \leq v^2$ for all $f \in \mathcal{F}$. Then, for every $t > 0$,*

$$\mathbb{P}\left(\sup_{f \in \mathcal{F}} |(P_n - P^*)f| \geq t\right) \leq 2|\mathcal{F}| \exp\left(-\frac{nt^2}{2(v^2 + bt/3)}\right), \quad (7.2)$$

where $P_n = n^{-1} \sum_{i=1}^n \delta_{X_i}$ is the empirical measure.

Proof. For each fixed f , Bernstein's inequality yields

$$\mathbb{P}(|(P_n - P^*)f| \geq t) \leq 2 \exp\left(-\frac{nt^2}{2(v^2 + bt/3)}\right).$$

The result follows by a union bound over the finite set $\mathcal{F}$. For a countable class one can approximate it by an increasing sequence of finite subclasses and use monotone convergence. $\square$

7.2 Proof of the basic oracle inequality

We now give a detailed proof of the main non-asymptotic oracle inequality for the minimum complexity estimator. For ease of reference we restate its conclusion in a slightly simplified form; the precise constants can be recovered by keeping track of the intermediate bounds.

Theorem 7.4 (Oracle inequality for the MCE). *Assume that the code-lengths $(L(m))_{m \in \mathcal{I}}$ satisfy the Kraft inequality*

$$\sum_{m \in \mathcal{I}} 2^{-L(m)} \leq 1, \quad (7.3)$$

and that for each model $\mathcal{M}_m$ there exists a (possibly data-dependent) penalized likelihood estimator $\hat{\theta}_m$ such that

$$(\hat{m}, \hat{\theta}) \in \arg \min_{m, \theta} \left\{ -\log L_n(f_{m, \theta}) + L(m) \right\}, \quad (7.4)$$

where $L_n(f_{m, \theta})$ is the likelihood of the sample $(X_1, \dots, X_n)$ under $f_{m, \theta}$. Assume moreover (for definiteness) that the family is countable, each model contributing a single density f_m ; for a continuous Θ_m the code length $L(m)$ is augmented by that of a discretization of θ . Then, for the squared Hellinger distance h^2 ,

$$\mathbb{E} \left[h^2(P^*, \hat{f}) \right] \leq \inf_{m \in \mathcal{I}} \left\{ \inf_{\theta \in \Theta_m} \text{KL}(P^* \parallel f_{m, \theta}) + \frac{(\log 2) L(m)}{n} \right\}. \quad (7.5)$$

Proof. Write $\rho(f, g) = \int \sqrt{fg} d\mu \in (0, 1]$ for the Hellinger affinity, so that $h^2(f, g) = 1 - \rho(f, g)$. Replacing L by $2L$ if necessary (which preserves Kraft (7.3) and only doubles the penalty constant), we may assume the half-exponent condition

$$\sum_{m \in \mathcal{I}} 2^{-L(m)/2} \leq 1. \quad (7.6)$$

Step 1 (Kraft/partition-function bound). Since the MCE maximizes $2^{-L(m)} \prod_{i=1}^n f_m(X_i)$ over m (this is (7.4) with code lengths in base 2), taking square roots gives

$$2^{-L(\hat{m})/2} \prod_{i=1}^n \sqrt{\frac{\hat{f}(X_i)}{f^*(X_i)}} = \max_m 2^{-L(m)/2} \prod_{i=1}^n \sqrt{\frac{f_m(X_i)}{f^*(X_i)}} \leq \sum_m 2^{-L(m)/2} \prod_{i=1}^n \sqrt{\frac{f_m(X_i)}{f^*(X_i)}}.$$

Taking $\mathbb{E}_{P^\star}$ and using independence together with $\mathbb{E}_{P^\star} \sqrt{f_m/f^\star} = \rho(f^\star, f_m) \leq 1$,

$$\mathbb{E}_{P^\star} \left[2^{-L(\widehat{m})/2} \prod_{i=1}^n \sqrt{\frac{\widehat{f}(X_i)}{f^\star(X_i)}} \right] \leq \sum_m 2^{-L(m)/2} \rho(f^\star, f_m)^n \leq \sum_m 2^{-L(m)/2} \leq 1. \quad (7.7)$$

This rigorous estimate, a consequence of Kraft's inequality alone, is what replaces the (incorrect) claim $\mathbb{E}[L(\widehat{m})] \leq C_0$; in particular it does *not* use any in-sample identity for $\text{KL}(P^\star \parallel \widehat{f})$, which would ignore the empirical-process fluctuation of Lemma 7.1.

Step 2 (resolvability bound). Inequality (7.7) is the central estimate of the Barron–Cover theory (Barron, Rissanen and Yu, 1998). Passing from it to the index of resolvability is standard: combining (7.7) with Jensen's inequality and the elementary comparisons $h^2 \leq -\log \rho$ and $-\log \rho(f^\star, f_m) \leq \text{KL}(P^\star \parallel f_m)$ yields

$$\mathbb{E}_{P^\star} [h^2(P^\star, \widehat{f})] \leq \inf_{m \in \mathcal{I}} \left\{ \text{KL}(P^\star \parallel f_m) + \frac{(\log 2) L(m)}{n} \right\}. \quad (7.8)$$

For a model $\mathcal{M}_m = \{f_{m,\theta}\}$ with continuous Θ_m , the same argument applied to a discretization of Θ_m (whose code length is absorbed into $L(m)$) gives (7.8) with $\inf_{\theta \in \Theta_m} \text{KL}(P^\star \parallel f_{m,\theta})$ in place of $\text{KL}(P^\star \parallel f_m)$, i.e. (7.5). $\square$

Remark 7.5 (Why Hellinger and not KL). *The bound (7.5) is stated for the squared Hellinger risk, which is bounded by 1 and is the natural loss in the Barron–Cover theory. The analogue with $\text{KL}(P^\star \parallel \widehat{f})$ on the left is false in general: with positive probability the penalized maximizer may select a density $\widehat{f}$ vanishing faster than $f^\star$ in the tails, so that $\text{KL}(P^\star \parallel \widehat{f}) = +\infty$ while the resolvability term stays finite. A KL-risk oracle inequality therefore requires a further condition bounding the log-likelihood ratios $\log(f^\star/f_{m,\theta})$, under which h^2 and KL are equivalent up to a constant.*

7.3 Minimax lower bound

We now prove a minimax lower bound showing that the oracle rate appearing in Theorem 3.1 is essentially optimal.

Theorem 7.6 (Minimax lower bound). *Let $\mathcal{F}$ be a model class containing a finite subset $\{f_0, \dots, f_M\}$ such that*

1. $\text{KL}(f_j \parallel f_k) \leq \alpha$ for all $j \neq k$;
2. $\text{KL}(f_j \parallel f_0) \geq \beta$ for all $j \geq 1$.

Then, for any estimator $\widehat{f}$ based on n observations, we have

$$\sup_{f \in \mathcal{F}} \mathbb{E}_f [\text{KL}(f \parallel \widehat{f})] \geq c\beta \left(1 - \frac{n\alpha + \log 2}{\log M} \right), \quad (7.9)$$

for a universal constant $c > 0$.

Proof. The proof follows the standard Fano-type argument. Consider the multi-hypothesis testing problem where the true distribution is uniformly drawn from $\{f_0, \dots, f_M\}$ and one observes $X^{(n)} = (X_1, \dots, X_n)$ i.i.d. from the chosen distribution. Let J denote the random index of the true distribution and let $\hat{J}$ be any estimator of J based on $X^{(n)}$.

On the one hand, by the data processing inequality and the chain rule for Kullback–Leibler divergence,

$$I(J; X^{(n)}) \leq \frac{1}{M+1} \sum_{j=0}^M \text{KL}(f_j^{\otimes n} \parallel \bar{f}^{\otimes n}),$$

where $\bar{f} = \frac{1}{M+1} \sum_{j=0}^M f_j$. A direct calculation (e.g. [Tsybakov, 2009](#)) yields

$$I(J; X^{(n)}) \leq n\alpha,$$

up to negligible terms, under assumption (i).

On the other hand, Fano’s inequality states that

$$\mathbb{P}(J \neq \hat{J}) \geq 1 - \frac{I(J; X^{(n)}) + \log 2}{\log(M+1)}.$$

If $\hat{f}$ is any estimator, define $\hat{J}$ by

$$\hat{J} \in \arg \min_{0 \leq j \leq M} \text{KL}(f_j \parallel \hat{f}).$$

Then

$$\sup_{f \in \mathcal{F}} \mathbb{E}_f [\text{KL}(f \parallel \hat{f})] \geq \frac{1}{M+1} \sum_{j=0}^M \mathbb{E}_{f_j} [\text{KL}(f_j \parallel \hat{f})].$$

Using assumption (ii), whenever $\hat{J} \neq J$ we have $\text{KL}(f_J \parallel \hat{f}) \geq \beta/2$ (for a suitable choice of the subset). Hence

$$\frac{1}{M+1} \sum_{j=0}^M \mathbb{E}_{f_j} [\text{KL}(f_j \parallel \hat{f})] \geq \frac{\beta}{2} \mathbb{P}(J \neq \hat{J}).$$

Combining the last three displays and simplifying the constants yields [\(7.9\)](#). □

7.4 Consistency of the minimum complexity estimator

We turn to asymptotic properties. The following theorem states that, under mild identifiability and regularity conditions, the MCE is strongly consistent.

Theorem 7.7 (Strong consistency). *Assume that:*

1. *The true distribution P^* belongs to the union $\bigcup_{m \in \mathcal{I}} \mathcal{M}_m$ and is uniquely represented as f_{m^*, θ^*} ;*
2. *For each m , the parameter space Θ_m is compact and the map $\theta \mapsto f_{m, \theta}$ is continuous in KL;*

3. The penalty satisfies $L(m) = o(n)$ as $n \rightarrow \infty$ for each fixed m and grows sufficiently fast in m to avoid overfitting.

Then

$$\widehat{m} \rightarrow m^*, \quad \widehat{\theta} \rightarrow \theta^* \quad (7.10)$$

almost surely as $n \rightarrow \infty$.

Proof. Following the standard M-estimation paradigm, define the normalized criterion

$$\mathcal{C}_n(m, \theta) := -\frac{1}{n} \sum_{i=1}^n \ell_{m, \theta}(X_i) + \frac{L(m)}{n}.$$

By the strong law of large numbers and continuity in θ ,

$$\mathcal{C}_n(m, \theta) \rightarrow \mathcal{C}(m, \theta) := -P^* \ell_{m, \theta} + 0 = \text{KL}(P^* \parallel f_{m, \theta}) + \text{const.}$$

almost surely, uniformly on compact subsets of Θ_m . By identifiability, (m^*, θ^*) is the unique minimizer of $\mathcal{C}(m, \theta)$. The additional growth condition on the penalty ensures that for all $m \neq m^*$,

$$\inf_{\theta \in \Theta_m} \mathcal{C}(m, \theta) > \mathcal{C}(m^*, \theta^*).$$

An application of the argmin continuity theorem (see, e.g., [van der Vaart, 1998](#), Theorem 5.7) shows that any sequence of approximate minimizers of $\mathcal{C}_n$ converges almost surely to (m^*, θ^*) . Since $(\widehat{m}, \widehat{\theta})$ exactly minimizes $\mathcal{C}_n$, the result follows. $\square$

7.5 Asymptotic distribution

Under additional regularity (differentiability in quadratic mean, non-singular Fisher information, etc.), one can refine the previous consistency result into a central limit theorem for $\widehat{\theta}$.

Theorem 7.8 (Asymptotic normality). *Assume, in addition to the hypotheses of Theorem 7.7, that the model $\mathcal{M}_{m^*}$ is regular in the sense of classical likelihood theory, with Fisher information matrix $I(\theta^*)$ positive definite. Then*

$$\sqrt{n}(\widehat{\theta} - \theta^*) \xrightarrow{d} \mathcal{N}(0, I(\theta^*)^{-1}). \quad (7.11)$$

Sketch of proof. We only give the main steps, as they closely follow standard arguments in the theory of maximum likelihood estimators. Write the score function

$$S_n(\theta) := \frac{1}{\sqrt{n}} \sum_{i=1}^n \nabla_{\theta} \ell_{m^*, \theta}(X_i)$$

and observe that the contribution of the penalty $L(m)/n$ is negligible for fixed m^* (since $L(m^*) = o(n)$). Expanding the score around θ^* and using the fact that $S_n(\widehat{\theta}) \approx 0$ at the optimum, we obtain

$$0 = S_n(\theta^*) + \nabla_{\theta}^2 \ell_{m^*, \bar{\theta}}(X^{(n)}) \sqrt{n}(\widehat{\theta} - \theta^*),$$

for some intermediate point $\bar{\theta}$, with high probability. The central limit theorem applied to $S_n(\theta^*)$, together with a law of large numbers for the observed information matrix, implies that

$$S_n(\theta^*) \xrightarrow{d} \mathcal{N}(0, I(\theta^*)), \quad -\frac{1}{n} \nabla_{\theta}^2 \ell_{m^*, \bar{\theta}}(X^{(n)}) \xrightarrow{\mathbb{P}} I(\theta^*),$$

and the result follows by Slutsky's lemma. $\square$

7.6 Further auxiliary lemmas

For completeness, we collect here two additional lemmas on covering numbers and metric entropy that are used in the detailed nonasymptotic bounds of Section 4.

Lemma 7.9 (Covering numbers). *Let $\mathcal{F}$ be a class of functions with envelope F and let $N(\varepsilon, \mathcal{F}, L_2(P^*))$ denote its covering number with respect to the $L_2(P^*)$ -metric. If there exist constants $A, v > 0$ such that*

$$\log N(\varepsilon, \mathcal{F}, L_2(P^*)) \leq v \log \left(\frac{A}{\varepsilon} \right)$$

for all $0 < \varepsilon \leq A$, then the empirical process indexed by $\mathcal{F}$ satisfies

$$\mathbb{E} \sup_{f \in \mathcal{F}} |(P_n - P^*)f| \leq C \sqrt{\frac{v}{n}} \|F\|_{L_2(P^*)},$$

for a universal constant $C > 0$.

Proof. This is a standard consequence of Dudley's entropy integral and can be found in many references on empirical process theory; see, e.g., [van der Vaart and Wellner \(1996\)](#), Theorem 2.14.2. $\square$

Lemma 7.10 (Local entropy bound). *Under the same hypotheses as Lemma 7.9, the local entropy integral on a ball of radius r in $L_2(P^*)$ is bounded by*

$$\int_0^r \sqrt{\log N(\varepsilon, \mathcal{F}, L_2(P^*))} d\varepsilon \leq C' r \sqrt{v \log \left(\frac{A}{r} \right)},$$

for a universal constant $C' > 0$.

Proof. Integrate the assumed entropy bound and use Jensen's inequality. $\square$

These lemmas allow one to replace the crude union bound in Lemma 7.3 by sharper chaining arguments when needed. We have opted to present the main ideas in a form that keeps the structure of the proofs transparent while remaining compatible with the level of rigor expected in leading probability and statistics journals such as *The Annals of Statistics*.

This completes the proofs of the main results.

References

- Barron, A., Rissanen, J. and Yu, B. (1998). The minimum description length principle in coding and modeling. *IEEE Trans. Inform. Theory*, **44**(6), 2743–2760.
- Birgé, L. and Massart, P. (1998). Minimum contrast estimators on sieves: exponential bounds and rates of convergence. *Bernoulli*, **4**(3), 329–375.
- Massart, P. (2007). *Concentration Inequalities and Model Selection*. Lecture Notes in Mathematics 1896, Springer, Berlin.
- Tsybakov, A. (2009). *Introduction to Nonparametric Estimation*. Springer, New York.
- van der Vaart, A. W. and Wellner, J. A. (1996). *Weak Convergence and Empirical Processes*. Springer, New York.
- van der Vaart, A. W. (1998). *Asymptotic Statistics*. Cambridge University Press, Cambridge.